



Strathmore
UNIVERSITY

FACULTY OF INFORMATION TECHNOLOGY

BBT 2104 – INTEGRAL CALCULUS

Date: 15th November 2016

Time: 2 Hours

Instructions

Answer QUESTION ONE and any other TWO questions

QUESTION ONE (30 Marks)

- a) Evaluate the following integral $\int \sin^3 x \cos^2 x \, dx$ [4 Marks]
- b) Determine :
- i. The area of the region enclosed by $y = -x^2 + 11$ and $y = x^2 + 5 - 4x$ [4 Marks]
- ii. The area of the region enclosed by the loops of the curve $y = 4x - x^3$ and the x axis [4 Marks]
- c) Evaluate using the method of change of variable $\int x (a^2 - x^2)^{0.5} \, dx$ where a is a constant. [3 Marks]
- d) By setting $x = 3\sin \theta$, evaluate $\int \frac{x^2 \, dx}{\sqrt{9-x^2}}$ [3 Marks]
- e) Use partial fractions to evaluate $\int \frac{x^2 + 4x + 1}{(x-1)(x+1)(x+3)} \, dx$ [4 Marks]
- f) Use Taylor's theorem to obtain a series expansion for $\cos(x + \frac{\pi}{3})$ up to and including terms in x^3 [4 Marks]
- g) Find the general solution of the differential equation $xy' + 2y' = y + 1$ [4 Marks]

QUESTION TWO (20 Marks)

- a) Determine the area of the region bounded by the curve $y = (x-1)(x-2)(x-3)$ and the lines $x = 0, x = 4$ as well as the x axis [5 Marks]
- b) Determine the area bounded by the curve $y^2 = 9 - x$ and the y axis [5 Marks]
- c) Derive the reduction formula for $\int \cos^n x \, dx$ hence evaluate $\int \cos^3 x \, dx$ [10 Marks]

QUESTION THREE (20 Marks)

Evaluate the following integrals using an appropriate method:

- i. $\int \sin 3x \cos 5x \, dx$ [5 Marks]
- ii. $\int x \sin x \, dx$ [5 Marks]
- iii. $\int x^2 e^x \, dx$ [5 Marks]
- iv. $\int e^x \cos x \, dx$ [5 Marks]

QUESTION FOUR (20 Marks)

- a) Find the volume of the solid generated by rotating about the x axis the region bounded by $y = x^2$, the x axis and the lines $x = 1$ and $x = 2$ [3 Marks]
- b) Suppose we have a cone of base radius r and vertical height h , imagine the cone being formed by rotating a straight line through the origin by an angle of 360 degrees about the x axis. Show that the volume of the cone is given by $\frac{\pi r^2 h}{3}$ [7 Marks]
- c) Solve the following differential equations
 - i. $3(y^2 + 1)dx + 2xy \, dy = 0$ [5 Marks]
 - ii. $y' = \frac{1}{x^2 y}$ [5 Marks]

QUESTION FIVE (20 Marks)

- a) Find the length of the arc $f(x) = \frac{x^3}{6} + \frac{x^{-1}}{2}$ from $x = 1$ to $x = 3$ [6 Marks]
- b) Find the mean with respect to x of the function $f(x) = 5x^2 - 4x$ [4 Marks]
- c) Find an approximate value for $\int x^2 e^x \, dx$ using the trapezium rule with 5 strips giving your answer correct to 4 significant figures. [6 Marks]
- d) Given that $y = f(x)$ and that for given values of x , the corresponding values for y are shown below. Use Simpson's rule with 8 strips to evaluate $\int f(x) \, dx$ [4 Marks]

x	0	1	2	3	4	5	6	7
y	10	5	2	1	2	5	10	17